Essential Summary

Quantitative Analysis of Dynamical Bifurcations in a Coupled Smooth and Discontinuous Oscillator with High-order Nonlinear Damping

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1 Background and Motivation

The coupled smooth and discontinuous (SD) oscillator, introduced by Han et al. (2012), consists of two SD oscillators coupled rigidly. Its restoring force contains **two irrational nonlinear terms**, which arise from the geometric configuration of oblique springs. These irrational terms make the system significantly harder to analyze than classical nonlinear oscillators.

When perturbed by high-order nonlinear damping of the form $(\mu_c + \mu_1 x + \mu_2 x^2 + \mu_3 x^3 + \mu_4 x^4)\dot{x}$, the oscillator exhibits rich dynamical behavior: limit cycles of varying stability, homoclinic and heteroclinic bifurcations, and multi-stable configurations (single-well, double-well, and triple-well potentials).

Classical quantitative methods—the elliptic function L–P method, the elliptic perturbation method, and the standard GHFP method—**cannot handle** this system because the two irrational terms render the key integrals analytically intractable. This work overcomes that barrier by extending the **modified generalized harmonic function perturbation (MGHFP)** method, originally developed by the authors for simpler oscillator classes, to this most challenging case.

2 System Model

The investigated oscillator is governed by:

$$\ddot{x} + (x+\beta)\frac{1}{\sqrt{(x+\beta)^2+\alpha^2}} + (x-\beta)\frac{1}{\sqrt{(x-\beta)^2+\alpha^2}} = \varepsilon(\mu_c + \mu_1 x + \mu_2 x^2 + \mu_3 x^3 + \mu_4 x^4)\dot{x} \quad (1)$$

where α is the smoothing parameter ($\alpha \to 0$ gives the non-smooth/discontinuous limit), β is the coupling parameter controlling the potential well structure, ε is a small positive perturbation parameter, μ_c is the control parameter, and μ_1–μ_4 are nonlinear damping coefficients.

The potential energy landscape undergoes qualitative changes as α and β vary: monostable (single-well) for large α/β ratios, bistable (double-well) at intermediate values, and tristable (triple-well) for small α with appropriate β.

3 The MGHFP Method — Key Steps

The method proceeds in three stages:

3.1 Stage 1: Nonlinear Time Transformation

Introduce a nonlinear time transformation $\frac{d\varphi}{dt} = \Phi(\varphi)$ with $\varphi(0) = 0$, $\varphi(t+\tau) = \varphi(t) + \pi$. The undamped system is transformed into:

$$\frac{d}{d\varphi}(\Phi x') + g(x) = 0, \qquad x(\varphi) = a\cos^2\Phi(\varphi) + b \quad (2)$$

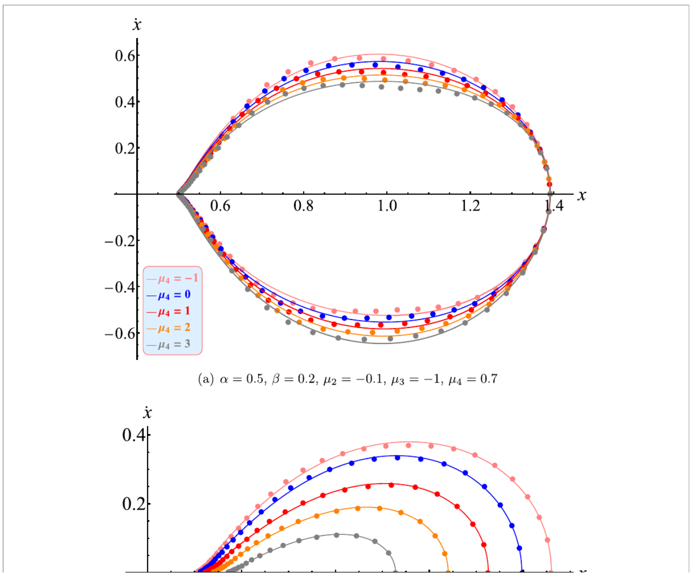


Figure 1: **Triple-well potential** of the coupled SD oscillator. The system has five equilibrium points: two centers, two saddles, and an additional center, creating a complex landscape where both homoclinic and heteroclinic orbits can coexist.

where $g(x) = (x+\beta)\left[1 - \frac{\alpha^2}{(x+\beta)^2+\alpha^2}\right] + (x-\beta)\left[1 - \frac{\alpha^2}{(x-\beta)^2+\alpha^2}\right]$.

The function $\Phi(\varphi)$ is expanded as a Fourier series:

$$\Phi_0(\varphi) = \sum_{i=0}^{m} (p_{2i} \cos 2i\varphi + q_{2i} \sin 2i\varphi) \tag{3}$$

3.2 Stage 2: Perturbation Expansion

The amplitude and frequency are expanded in powers of ε:

$$a = a_0 + \varepsilon a_1 + \cdots, \qquad \Phi = \Phi_0 + \varepsilon \Phi_1 + \cdots \tag{4}$$

Substituting into the damped system and collecting terms at each order of ε yields a hierarchy of equations. The zeroth-order gives the unperturbed solution; the first-order determines a_1 and Φ_1.

3.3 Stage 3: Composite Simpson Integration — The Key Innovation

The Fourier coefficients p_{2i} involve integrals of the irrational nonlinear terms that **cannot be evaluated analytically**. The MGHFP method's key innovation is the introduction of the **composite Simpson quadrature formula** to discretize these integrals into sums over equally-spaced nodes, making the coefficients computable to high precision.

Specifically, the integration interval $[0, \pi]$ is divided into eight equal parts, and the Simpson formula yields explicit expressions for each p_{2i} in terms of a_0, b, α, and β (listed in Appendix A of the original paper).

4 Key Equations

4.1 Amplitude–Parameter Relation (Central Result)

Setting the Fourier truncation order $m = 4$, the analytical relationship between the limit cycle amplitude A and the control parameter μ_c is:

$$\boxed{\mu_c = -\frac{4}{a_0^2(2p_0 - p_4)}\Big(E_1\mu_1 + E_2\mu_2 + E_3\mu_3 + E_4\mu_4\Big)} \tag{5}$$

where p_{2i} are the Fourier coefficients computed via composite Simpson integration, and E_1–E_4 are explicit functions of a_0, b, and p_{2i}. **This is the central result**: given system parameters, the limit cycle amplitude is obtained directly—no numerical integration needed.

For symmetric limit cycles: $a_0 = 2A$, $b = -A$. For asymmetric limit cycles (triple-well case): $a_0 = A - C$, $b = C$, where C depends on α and β.

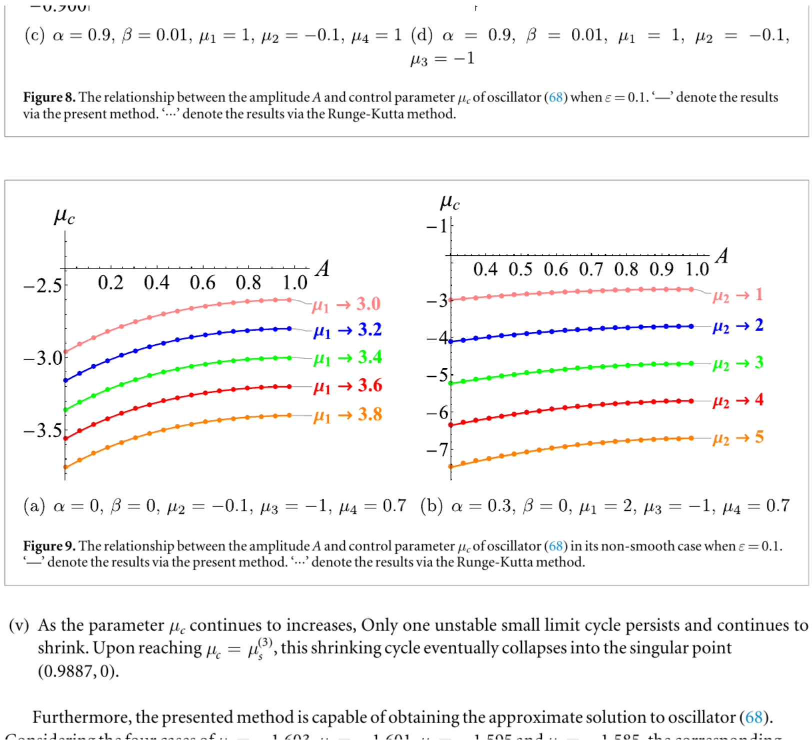


Figure 2: **Figure 1.** (Upper) Relationship between limit cycle amplitude A and control parameter μ_c for the single-well oscillator ($\alpha = 2$, $\beta = 0.5$). (Lower) Characteristic quantity h_0 determining stability. Key bifurcation points: $\mu^{(0)} = -0.1249$ (semi-stable cycle emerges), $\mu^{(1)} = -0.05$ (split into stable + unstable), $\mu^{(2)} = 0.02$ (unstable cycle collapses). The complete lifecycle of limit cycles is quantitatively captured.

4.2 Stability Criterion

The stability of each limit cycle is determined by the characteristic quantity:

$$\boxed{h_0 = \frac{1}{\pi}\int_0^{\pi}\left[\mu_c + \sum_{i=1}^{4}\mu_i(a_0\cos^2\varphi + b)^i\right]d\varphi} \tag{6}$$

Based on the qualitative theory of ODEs (Dumortier, Llibre & Artés, 2006):

- $h_0 > 0$: the limit cycle is **unstable**
- $h_0 = 0$: the limit cycle is **semi-stable** (a bifurcation point)
- $h_0 < 0$: the limit cycle is **stable**

4.3 Homo-heteroclinic Bifurcation Thresholds

Setting $b = h$ (saddle point ordinate) in Eq. (5) yields the critical parameter μ_c^{hom} for **homoclinic** bifurcation. Setting $b = h_2$ and $a_0 + b = h_1$ (connecting two saddle points) yields μ_c^{het} for **heteroclinic** bifurcation. For the triple-well system, both can coexist at the same saddle—a scenario uniquely handled by this method.

4.4 First-Order Approximate Solution

The limit cycle and its derivative are approximated as:

$$x = 2A\cos^2\Phi - A + \mathcal{O}(\varepsilon^2) \tag{7}$$

$$\dot{x} = -2a_0[\Phi_0 + \varepsilon\Phi_1]\cos\varphi\sin\varphi + \mathcal{O}(\varepsilon^2) \tag{8}$$

where $\Phi_0 = p_0 + p_2\cos 2\varphi + p_4\cos 4\varphi + p_6\cos 6\varphi + p_8\cos 8\varphi + \cdots$ and $\Phi_1 = l_2\sin 2\varphi + l_4\sin 4\varphi + l_6\sin 6\varphi + l_8\sin 8\varphi + \cdots$

5 Validated Examples

5.1 Example 1: Single-Well Potential ($\alpha = 2$, $\beta = 0.5$)

With $\mu_1 = 1$, $\mu_2 = 1$, $\mu_3 = 1$, $\mu_4 = -1$, the amplitude–parameter curve (Figure 1) reveals:

- $\mu_c < \mu^{(0)} = -0.1249$: **no limit cycle exists**.
- At $\mu_c = \mu^{(0)}$: a **semi-stable** limit cycle emerges (outer-stable, inner-unstable).
- $\mu^{(0)} < \mu_c < 0$: the semi-stable cycle **splits into two**—a smaller unstable cycle ($h_0 > 0$) and a larger stable cycle ($h_0 < 0$).
- At $\mu_c = 0$: the unstable cycle **collapses** to the singular point.
- $\mu_c > 0$: only a single **stable** limit cycle persists, growing with μ_c.

Accuracy metrics (Maximum Absolute Error in Amplitude, RMS Phase Error, and L_2-norm difference) confirm reliability: when $|\varepsilon\mu_i| \leq 0.5$, all errors remain below 5%.

5.2 Example 2: Triple-Well Potential

The triple-well case is considerably more complex: the system has **five equilibrium points**, and both homoclinic and heteroclinic orbits can coexist. The MGHFP method successfully:

- Predicts the critical values of μ_c for both homoclinic and heteroclinic bifurcations.
- Obtains approximate analytical solutions for the homo-heteroclinic orbits themselves.
- Tracks the evolution of both small limit cycles (enclosed by homoclinic orbits) and large limit cycles (enclosing all equilibrium points).

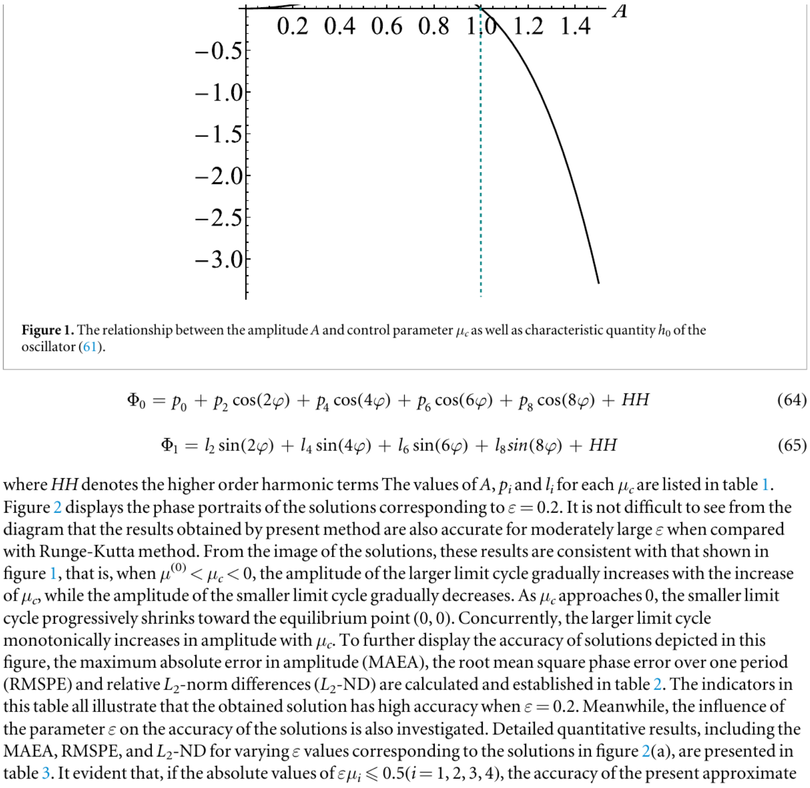


Figure 1. The relationship between the amplitude A and control parameter μ_c as well as characteristic quantity h_0 of the oscillator (61).

$$\Phi_0 = p_0 + p_2\cos(2\varphi) + p_4\cos(4\varphi) + p_6\cos(6\varphi) + p_8\cos(8\varphi) + HH \tag{64}$$

$$\Phi_1 = l_2\sin(2\varphi) + l_4\sin(4\varphi) + l_6\sin(6\varphi) + l_8 sin(8\varphi) + HH \tag{65}$$

where HH denotes the higher order harmonic terms The values of A, p_i and l_i for each μ_c are listed in table 1. Figure 2 displays the phase portraits of the solutions corresponding to $\varepsilon = 0.2$. It is not difficult to see from the diagram that the results obtained by present method are also accurate for moderately large ε when compared with Runge-Kutta method. From the image of the solutions, these results are consistent with that shown in figure 1, that is, when $\mu^{(0)} < \mu_c < 0$, the amplitude of the larger limit cycle gradually increases with the increase of μ_c, while the amplitude of the smaller limit cycle gradually decreases. As μ_c approaches 0, the smaller limit cycle progressively shrinks toward the equilibrium point (0, 0). Concurrently, the larger limit cycle monotonically increases in amplitude with μ_c. To further display the accuracy of solutions depicted in this figure, the maximum absolute error in amplitude (MAEA), the root mean square phase error over one period (RMSPE) and relative L_2-norm differences (L_2-ND) are calculated and established in table 2. The indicators in this table all illustrate that the obtained solution has high accuracy when $\varepsilon = 0.2$. Meanwhile, the influence of the parameter ε on the accuracy of the solutions is also investigated. Detailed quantitative results, including the MAEA, RMSPE, and L_2-ND for varying ε values corresponding to the solutions in figure 2(a), are presented in table 3. It evident that, if the absolute values of $\varepsilon\mu_i \leqslant 0.5 (i = 1, 2, 3, 4)$, the accuracy of the present approximate

Figure 3: **Figure 2.** Phase portraits of limit cycles for the single-well oscillator at different μ_c values ($\varepsilon = 0.2$). Solid lines: Runge–Kutta numerical integration. Dotted lines: present MGHFP analytical method. The excellent agreement confirms accuracy even at moderately large ε.

6 Key Findings

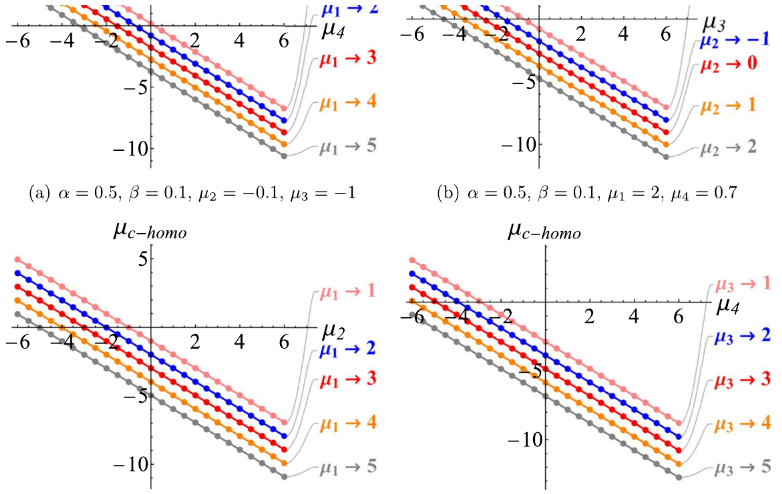
1. **Full quantitative prediction:** The MGHFP method achieves, for the first time, analytical quantitative analysis of limit cycles in a coupled SD oscillator with two irrational nonlinearities. Given system parameters, one can directly compute the number, amplitude, and existence interval of every limit cycle—no repetitive numerical search needed.

2. **Analytical stability criterion:** The characteristic quantity h_0 distinguishes stable, unstable, and semi-stable limit cycles analytically. Unstable and semi-stable limit cycles, which are exceedingly difficult to capture by purely numerical means, are reliably identified by this method.

3. **Simultaneous homo-heteroclinic bifurcation prediction:** For the triple-well system, homoclinic and heteroclinic orbits coexist at the same saddle. The MGHFP method predicts both bifurcation parameters simultaneously—a capability not available from prior analytical approaches.

4. **Complete limit cycle lifecycle:** The entire lifecycle is quantitatively tracked—from birth (semi-stable bifurcation), through evolution (splitting into stable + unstable pair), to annihilation (shrinking to a singular point).

5. **High-order damping applicability:** With quartic nonlinear damping, the method maintains reliable accuracy when $|\varepsilon\mu_i| \leq 0.5$. Beyond this range, accuracy is recovered by increasing the Fourier truncation order.



(a) $\alpha = 0.5,\ \beta = 0.1,\ \mu_2 = -0.1,\ \mu_3 = -1$

(b) $\alpha = 0.5,\ \beta = 0.1,\ \mu_1 = 2,\ \mu_4 = 0.7$

(c) $\alpha = 0.9,\ \beta = 0.01,\ \mu_3 = -1,\ \mu_4 = 1$

(d) $\alpha = 0.9,\ \beta = 0.01,\ \mu_1 = 2,\ \mu_2 = -0.1$

Figure 12. The variation of critical value of heteroclinic bifurcation parameter of oscillator (68) with system parameter $\mu_{c-hetero}$ when $\varepsilon = 0.1$. '—' denote the results via the Runge-Kutta method. '···' denote the results via the present method.

Table 10. The values of a_0, b, p_i and l_i for each β in homoclinic solutions which shown in figure 11(b).

β	0.2	0.24	0.28	0.32	0.36
μ_c	0.00142829	0.00136716	0.00125627	0.00104343	0.0005722
a_0	−1.0128	−1.0250	−1.0455	−1.0803	−1.1437
b	0.5064	0.5125	0.5228	0.5402	0.5719
p_0	0.2915	0.2590	0.2301	0.2034	0.1761
p_2	0.0055	0.0047	0.0040	0.0034	0.0027
p_4	0.2029	0.1866	0.1709	0.1561	0.1420
p_6	0.0051	0.0043	0.0037	0.0031	0.0025
p_8	−0.0485	−0.0387	−0.0306	−0.0235	−0.0159
l_2	−0.1175	−0.1166	−0.1166	−0.1177	−0.1201
l_4	−0.0008	−0.0008	−0.0007	−0.0005	−0.0001
l_6	0.0068	0.0060	0.0053	0.0048	0.0042
l_8	0.0002	0.0002	0.0002	0.0003	0.0004

Figure 4: **Figure 7.** Limit cycles for the triple-well oscillator at different μ_c values ($\varepsilon = 1$). Despite the large perturbation parameter, the MGHFP results (dotted) closely match the Runge–Kutta reference (solid), demonstrating the method's robustness in complex multi-stable systems.

7 Method Limitations

The authors identify three conditions where accuracy degrades:

- *Large perturbation parameter* ($\varepsilon > 0.5$): Relative error exceeds 5%. Extending to higher-order perturbation expansions can mitigate this.
- *Limit cycles crossing saddle points*: The velocity drops sharply near the saddle, causing abrupt displacement waveform features. The Fourier series truncated at $m = 4$ may be insufficient; higher truncation orders are needed.
- *Non-smooth potential wells* ($\alpha = 0$): The non-smoothness affects accuracy. A smaller ε or higher truncation order is recommended.

8 Applications and Significance

The mathematical model investigated—combining irrational nonlinear stiffness, high-order nonlinear damping, and multi-stable dynamics—has direct relevance to two engineering domains:

(1) Quasi-zero stiffness (QZS) vibration isolators: Recent research has extensively explored QZS isolators incorporating nonlinear electromagnetic damping for low-frequency vibration attenuation. The coupled SD oscillator's irrational stiffness and high-order damping naturally model such systems, and the MGHFP method provides a quantitative tool for predicting their limit cycle behavior and bifurcation thresholds.

(2) MEMS resonators: In micro-electro-mechanical systems, nonlinear damping and multi-stable dynamics are increasingly exploited for performance enhancement. The bifurcation analysis framework developed here directly supports the design and optimization of such devices, where saddle-node bifurcations and multi-stability play critical roles.

Beyond these specific applications, the MGHFP method itself constitutes a versatile analytical tool: it can be transferred to other systems with irrational or non-smooth nonlinearities, wherever classical perturbation methods fail.

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